

Daniel Alfredo Barreras Meraz

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EDUCATION

Tecnológico de Monterrey (ITESM)

B.S. in Computer Science and Technology, GPA: 98/100

Guadalajara, MX

Expected Graduation: Dec. 2026

EXPERIENCE

Lead Software Engineer, Full Stack

Jun. 2026 – [Incoming]

Capital One, Enterprise Platforms Technology

Mexico City, MX

- Lead a team of engineers building distributed microservices and full-stack systems across Capital One's enterprise platforms, delivering cloud-native AWS solutions serving millions of customers.
- Drive technical direction and mentor engineers, collaborating with product managers to translate regulatory and customer needs into scalable architecture.
- Design distributed event-driven systems using Java, Kafka, and Kinesis for high-throughput financial data processing at enterprise scale.
- Spearhead adoption of container orchestration (Docker, Kubernetes) and open-source frameworks to modernize deployment workflows and accelerate delivery cycles.

Research Engineer

Jul. 2025 – Present

Harvard Medical School – Boston Children's Hospital (FNNDSC)

Boston, MA

- Designed and trained FetEnhNet, a tissue-conditioned deep learning architecture for fetal MRI enhancement, achieving +4.48 dB PSNR across 10,000+ volumetric samples from 6 global sites.
- Built production-scale ETL pipeline reducing batch processing time 75% via parallel processing and Docker containerization, under strict HIPAA/IRB compliance.
- Developed ActivSparse-SLM, a 125M sparse LLM combining SLTrain, DeepSeekMoE, and Wanda pruning — 50% FFN weight sparsity with only +0.3% perplexity degradation.
- Adapted IGUANe CycleGAN from adult T1 to fetal T2-weighted MRI for cross-site scanner harmonization across 6 global acquisition protocols.

Sr Full Stack Engineer

Nov. 2023 – Dec. 2025

Sidepocket Inc

San Francisco, CA (Remote)

- Scaled fintech microservices from 2K to 15K users handling 2M+ daily transactions at sub-200ms latency, maintaining 99.9% uptime and cutting infrastructure costs 30%.
- Eliminated 85% CI/CD manual overhead via infrastructure-as-code on AWS ECS; mentored 3 engineers through rapid team growth.
- Architected Redis-to-S3 Parquet migration across 8 microservices, replacing ElastiCache for cost reduction and improved query flexibility.
- Designed server-side filtered cache endpoints reducing wire transfer from 60MB to matching rows only, improving high-frequency portfolio query performance.

Database Administrator

Jun. 2024 – Aug. 2025

Ciena Corp

Zapopan, MX

- Developed Python-based workflow automation with AWS CloudWatch integration, eliminating 70% manual operations while processing 10M+ daily records across Oracle EBS infrastructure.
- Accelerated mission-critical PL/SQL queries 45% (2.1s to 800ms) by applying indexing strategies, execution plan analysis, and query refactoring across distributed databases.
- Designed real-time database health monitoring and alerting pipelines via AWS CloudWatch, enabling proactive detection of performance regressions across critical Oracle EBS modules.

RESEARCH & ML PROJECTS

Phoenix Eye: AI Wildfire Detection | Python, TensorFlow, Next.js, Mapbox, Azure

Sep. 2023 – Dec. 2024

- Trained a multi-class CNN on 50,000+ MODIS/Landsat satellite images achieving 95% wildfire detection accuracy; integrated real-time geospatial inference with Mapbox GL 3D terrain rendering and time-control playback on a Next.js frontend.
- Deployed end-to-end on Azure with real-time alert pipelines; recognized as NASA Space Apps Global Finalist (Top 40 / 5,000+ teams) and Dubai Prototypes for Humanity 2024 Finalist.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript/TypeScript, C++, Go, Rust, PL/SQL, MATLAB, R, Scala

ML & AI: PyTorch, TensorFlow, Computer Vision, Deep Learning, Geospatial Analysis, NLP, scikit-learn, nibabel

Backend: Spring Boot, Flask, FastAPI, Microservices, Distributed Systems, RESTful APIs, GraphQL, ZeroRPC

Cloud & DevOps: AWS (ECS, Lambda, RDS, S3, CloudWatch, EC2), Docker, Kubernetes, CI/CD, Terraform, OCI

Databases: PostgreSQL, MongoDB, Oracle (EBS, PL/SQL), DynamoDB, Redis, Workflow Automation, ETL Pipelines

Frontend: React, Next.js, Vue.js, D3.js, Mapbox, Geospatial Visualization, HTML5/CSS3